

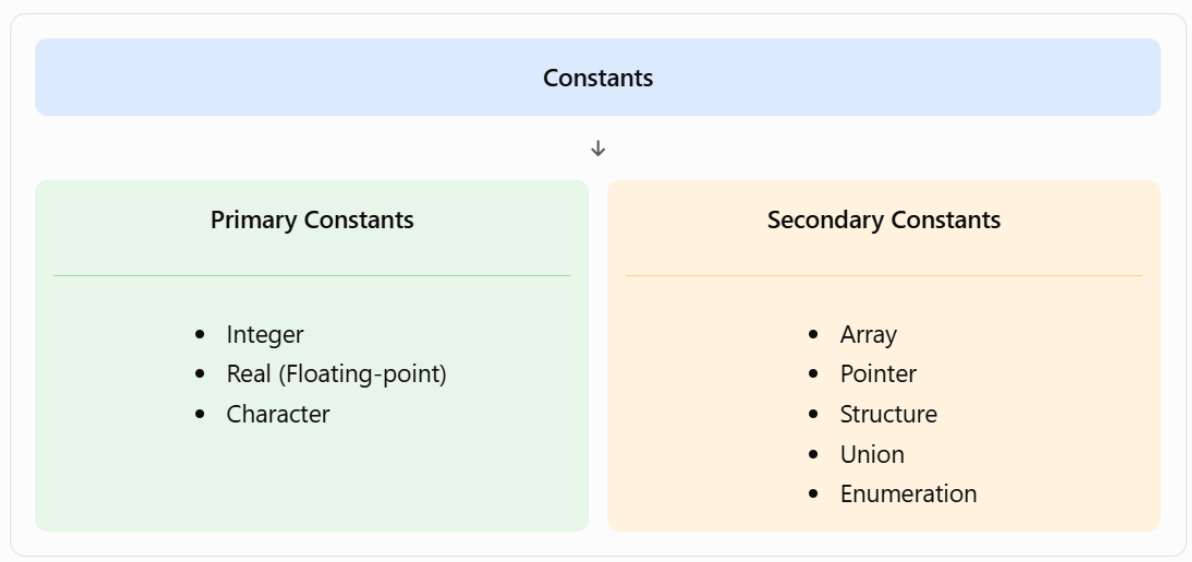
Constants

A constant is an entity that doesn't change whereas a variable is an entity that may change

X=3

X=5

Types of C Constants



Integer Constants

Integer constants are whole numbers without decimal points.

Rules for Constructing Integer Constants

- (a) An integer constant must not have a decimal point.
- (b) It can be either positive or negative.
- (c) If no sign precedes an integer constant it is assumed to be positive.
- (d) No commas or blanks are allowed within an integer constant.
- (e) It must have at least one digit

426

+782

-8000

-7605

Floating-Point Constants

Floating-point constants contain a **decimal point** or are represented using **exponential notation**.

Rules for Constructing Real Constants

Following rules must be observed while constructing real constants expressed in **fractional form**:

- (a) A real constant must have at least one digit.
- (b) It must have a decimal point.

- (c) It could be either positive or negative.
- (d) Default sign is positive.
- (e) No commas or blanks are allowed within a real constant.

+325.34
426.0
-32.76
-48.5792

Following rules must be observed while constructing real constants expressed in **exponential form**:

- (a) The **mantissa part and the exponential part** should be separated by a letter e.
- (b) The mantissa part may have a positive or negative sign.
- (c) Default sign of mantissa part is positive.
- (d) The exponent must have at least one digit, which must be a positive or negative integer. Default sign is positive.

+3.2e-5
4.1e8
-0.2e+3
-3.2e-5
1.5e3 [1.5×10³ =1500]

Character Constants

A character constant represents a single character enclosed in single quotation marks.

Rules for Constructing Character Constants

A character constant is a single alphabet, a single digit or a single special symbol enclosed within single inverted commas.

The maximum length of a character constant can be 1 character.

'A'
'I'
'5'
'='

String Literals

A **string literal** is a sequence of characters enclosed in double quotation marks

"Hello"
"Anna University"
"C Programming"

A string literal is terminated by a null character ('\0') internally.

Data Type	Size	Signed Range	Unsigned Range
char	1 byte	-128 to 127	0 to 255
short int	2 bytes	-32,768 to 32,767	0 to 65,535
int	4 bytes	-2,147,483,648 to 2,147,483,647	0 to 4,294,967,295
long int	8 bytes	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	0 to 18,446,744,073,709,551,615
float	4 bytes	Approximately $\pm 3.4 \times 10^{38}$	—
double	8 bytes	Approximately $\pm 1.7 \times 10^{308}$	—

Data Type	Example Declaration
signed char	signed char a = -100;
unsigned char	unsigned char b = 200;
short int	short int c = -30000;
unsigned short int	unsigned short int d = 60000;
signed int	signed int e = -200000;
unsigned int	unsigned int f = 400000;
long int	long int g = -500000;
unsigned long int	unsigned long int h = 1000000;